

A quantitative PCR method for *in planta* fungal DNA quantification in chickpea roots

1. Materials

1.1 Biological tissue samples

1. Host Material: Chickpea root samples collected from infected samples and control treatments.
2. Pathogen substrate: Clean fungal mycelium harvested via filter mesh from an actively growing 5-day-old, 250 mL Potato Dextrose Broth (PDB) culture of *Macrophomina phaseolina*.

1.2 Reagents

1. Cetyltrimethylammonium bromide (CTAB) extraction buffer components (see Note 1).
2. Liquid nitrogen.
3. Microcentrifuge tube
4. NanoDrop Spectrophotometer (Thermo Fisher Scientific, Waltham, MA, USA).
5. Real-Time quantitative PCR (qPCR) thermal cycler.

2. Methods

In-planta fungal DNA quantification

2.1. DNA Isolation using CTAB method

1. Take 250 mg of frozen root sample and add 2 mL of CTAB extraction buffer.
2. Incubate at 70°C for 1 hr and cool at room temperature.
3. Prepare chloroform: Isoamyl alcohol in a ratio of 24:1 and add equal volume of this to all the tubes.
4. Mix the tubes well and centrifuge at 10,000g for 10 mins at room temperature.

5. Transfer the aqueous phase to a fresh tube without disturbing, add 0.6 volume of isopropanol.
6. Centrifuge the tubes at 10,000g for 10 mins at 4°C.
7. Discard the supernatant, wash the pellet with 70% ethanol, and centrifuge at 10,000g for 10 mins at 4°C.
8. Air dry the pellet and dissolve it in 30 µL TE buffer.
9. Measure the quality and quantity of genomic DNA using 0.8% agarose gel and NanoDrop spectrophotometer.
10. Confirm the presence of *M. phaseolina* using PCR with specific primers (see Note 2).

2.2 Quantification of M. phaseolina using q PCR

2.2.1 DNA concentration normalization

1. Normalize the DNA concentration to 50 ng/µL.

2.2.2 Generation of standard calibration curve

1. Collect the fungal mycelia from a 5-day old fungal plate and isolate fungal genomic DNA using the protocol mentioned above.
2. Dilute the pure fungal genomic DNA to tenfold serially (from 100 to 1 pg/µl) with sterile autoclaved MQ.
3. Prepare the master mix of 9 µL containing 5 µL KAPA SYBR FAST Master Mix (2X) ABI Prism™ (Kapa Biosystems, Woburn, MA, USA) and 1 µL of each of the primers (Mph-F and Mph-R) (450 nm) and 3 µL nuclease-free water.
4. Aliquot 9 µL master mix to all the wells of the 96 well PCR plate.
5. Add 1 µL of genomic DNA of *M. phaseolina* to the plate, from 1 pg dilution to 100 ng dilution for standard curve preparation. Include three replicates per dilution.

6. Add 1 μL of DNA isolated from the root sample (50 ng/ μL) to each well separately (See Note 3).
7. Add 1 μL water to three wells for the No Template Control (NTC).
8. Load the 96 well PCR plate containing samples into the 7900HT Fast RT-qPCR System.
9. Adjust the program to include an initial denaturation step for 5 min at 95°C followed by 40 cycles of denaturation at 95°C for 10 s and annealing at 65°C for 35 s followed by heating to 95°C, cooling to 65°C for 1 min, and then heating to 97°C at a rate of 1°C/5 s.
10. Report the results as the mean C q value. Additionally, the target can be reported as fungal concentration per defined unit of the template.
11. Determine the concentration of fungal DNA present from the linear equation of the amplification using the target fungal gene C q values.
12. Generate the linear equation from the standard curve of known starting DNA concentrations containing the target gene (see Note 4).

4. Notes

1. CTAB Extraction Optimization: Chickpea root tissues contain high levels of polyphenols and polysaccharides that can co-precipitate with nucleic acids, potentially inhibiting downstream qPCR assays. Adding 2% (w/v) polyvinylpyrrolidone (PVP) directly into the initial CTAB extraction buffer helps bind and eliminate these contaminants.
2. Mph-F – AAACACACTTTTCGCACTCCTGC; Mph-R – TATGCTCGCAGAGAAGAACACGA are specific primers for the confirmation of *M. phaseolina*.

3. Ensure no amplification with target primers in the control plant samples or the NTC that indicate the absence of *M. phaseolina* DNA in the uninfected roots and no cross-contamination or primer dimers.

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Irulappan et al (2022) Drought Stress Exacerbates Fungal Colonization and Endodermal Invasion and Dampens Defense Responses to Increase Dry Root Rot in Chickpea. *Molecular Plant-Microbe Interactions*, 35(7), 583-591.

Mirchandani, R., Kandpal, M., Ranjan, A., Sinharoy, S., & Senthil-Kumar, M. (2025). Induced post-invasive defenses in the nonhost plant *Parthenium hysterophorus* L. prevent root cortical colonization by *Macrophomina phaseolina* and impart resistance to dry root rot. *Environmental and Experimental Botany*, 237, 106197.